BYoung Physical Therapy

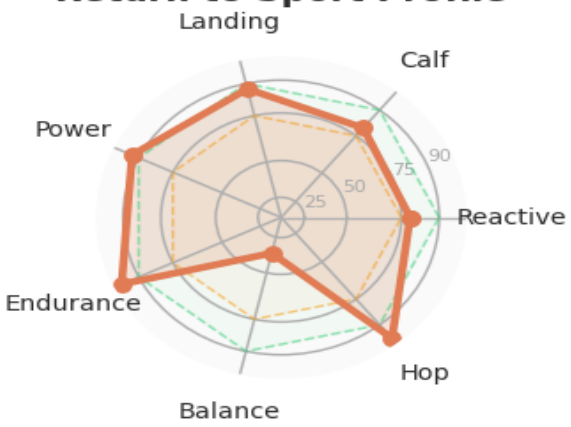
Patient: John Jiles • **Date:** November 09, 2026
Surgical Side: Left • **Surgery:** 04/21/2026 (6.6 mo post-op) • **Triple Hop:** Surg 22'0" / NS 22'7" / LSI 97.4%

Return to Sport Assessment
Achilles Repair

38

RTS Score
Not Ready
6 scored red flags · 6.6 mo post-op

Return to Sport Profile



▲ STRENGTHS

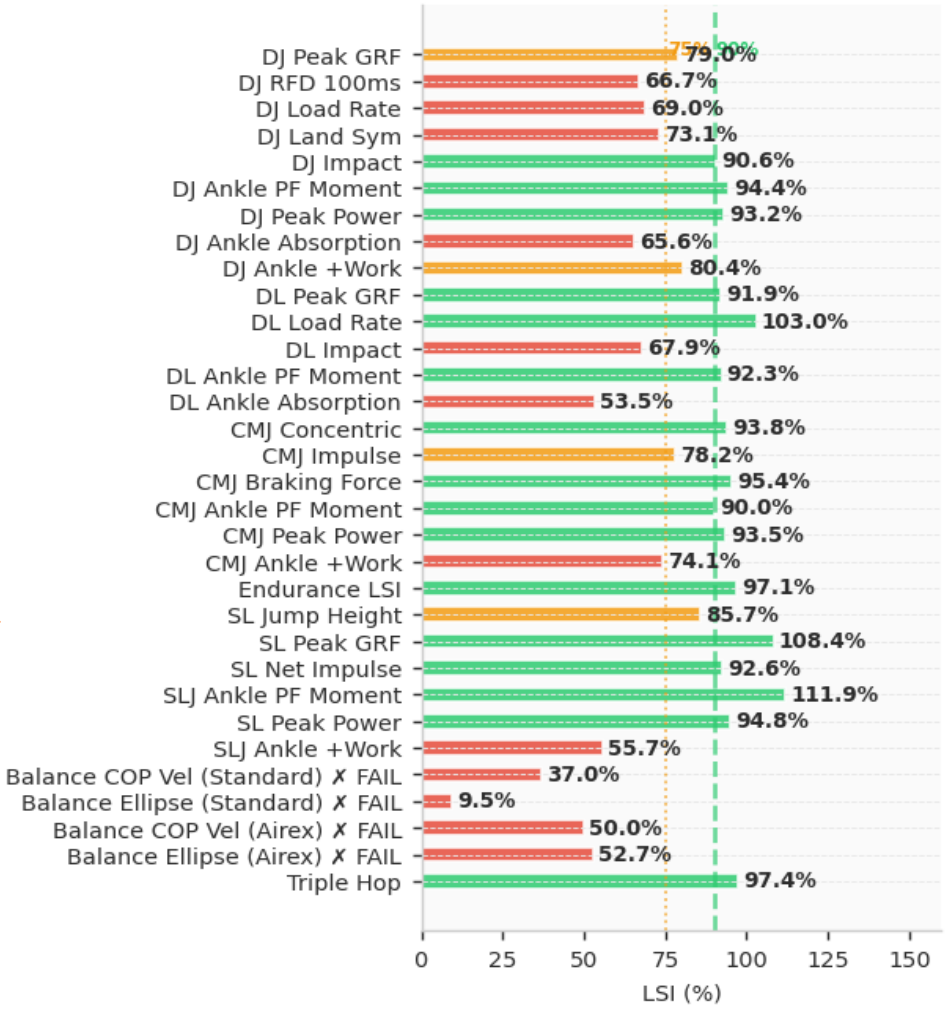
- Ankle PF moment sym. (100/100)
- DL Load rate sym. (100/100)
- Fatigue drift (100/100)
- Endurance symmetry (91/100)

▼ DETRACTORS

- Balance symmetry (30/100)
- Balance symmetry (Airex) (37/100)
- Ankle push-off work (SLJ) (40/100)
- Landing symmetry (48/100)
- Rate of force dev. (48/100)

Performance 65/100 → ×0.71 red-flag penalty (6 scored) → ×0.82 time modifier → = 38/100

Limb Symmetry Index — All Tests



CLINICAL FLAGS

- Balance (Airex): FAILED stance hold — surgical (foot moved), non-surgical (foot moved)
- Balance (Standard): FAILED stance hold — surgical (foot moved + touchdown), non-surgical (touchdown)
- Balance Ellipse (Standard) ✗ FAIL: LSI 9.5% — significant asymmetry
- Balance COP Vel (Standard) ✗ FAIL: LSI 37.0% — significant asymmetry
- SLJ Ankle Dorsiflexion: Δ -8.8° — exceeds 6° threshold

• +13 more — see individual test pages.

Body weight 212 lb from the plates vs 190 lb in the Visual3D session — plate value used; fix the session weight (it scales joint moments).

● ≥90% Symmetric ● ≥75% Progressing ● <75% Deficit LSI = Surgical ÷ Non-Surgical × 100 | Δ = Surgical – Non-Surgical (absolute difference; used for small-angle kinematics)

CALF & ANKLE — ACHILLES CRITERIA

2 of 13 Achilles criteria met · heel raise not collected (score uses the remaining criteria)

Criterion	Surgical	Non-Surg	LSI	Target	Status
CALF CAPACITY					
Single-leg heel raise (reps) *(not collected)*	—	—	—	LSI ≥ 90%	NOT TESTED
SL jump — ankle push-off work	90 J	162 J	56%	LSI ≥ 90%	DEFICIT
SL jump — peak plantarflexor moment	264 Nm	236 Nm	112%	LSI ≥ 90%	PASS
SINGLE-LEG POWER					
SL jump height	11.9 in	13.9 in	86%	LSI ≥ 90%	PROGRESSING
SL jump peak power	45.2 W/kg	47.7 W/kg	95%	LSI ≥ 90%	PASS
CMJ — ankle push-off work	85 J	115 J	74%	LSI ≥ 90%	DEFICIT
REACTIVE STRENGTH					
Drop jump RSI (bilateral)	1.26 m/s		—	≥ 1.30 m/s	PROGRESSING
Drop jump contact time	0.48 s		—	< 0.25 s	DEFICIT
Drop jump RFD (100 ms)	19,667 N/s	29,484 N/s	67%	LSI ≥ 85%	DEFICIT
Drop jump loading rate (50 ms)	17,308 N/s	25,072 N/s	69%	LSI ≥ 85%	DEFICIT
LANDING CONTROL					
Drop landing — ankle absorption	50 J	93 J	53%	LSI ≥ 90%	DEFICIT
Drop landing — impact transient	708 N	1,043 N	68%	LSI ≥ 90%	DEFICIT
BALANCE					
Balance — eyes closed, firm *(both limbs failed)*	FAIL	FAIL	37%	hold + LSI ≥ 90%	FAIL
Balance — eyes closed, Airex *(both limbs failed)*	FAIL	FAIL	71%	hold + LSI ≥ 90%	FAIL
HOP					
Triple hop for distance *(hop distance normalises early)*			97%	LSI ≥ 90%	PASS

WHAT THE CALF IS DOING

The force is there, the push-off isn't. On the single-leg jump the surgical calf reaches 112% of the other side's peak plantarflexor moment but does only 56% of its push-off work. It can produce the moment, but it isn't driving the leg through most of the push-off (see the power curve below) — strength isn't the limiter, using it quickly through range is.

The hip is covering for the calf. Hip push-off work is 91% of the other side while the ankle is 56% — a hip-dominant strategy. Height and power look closer to symmetric than the calf really is.

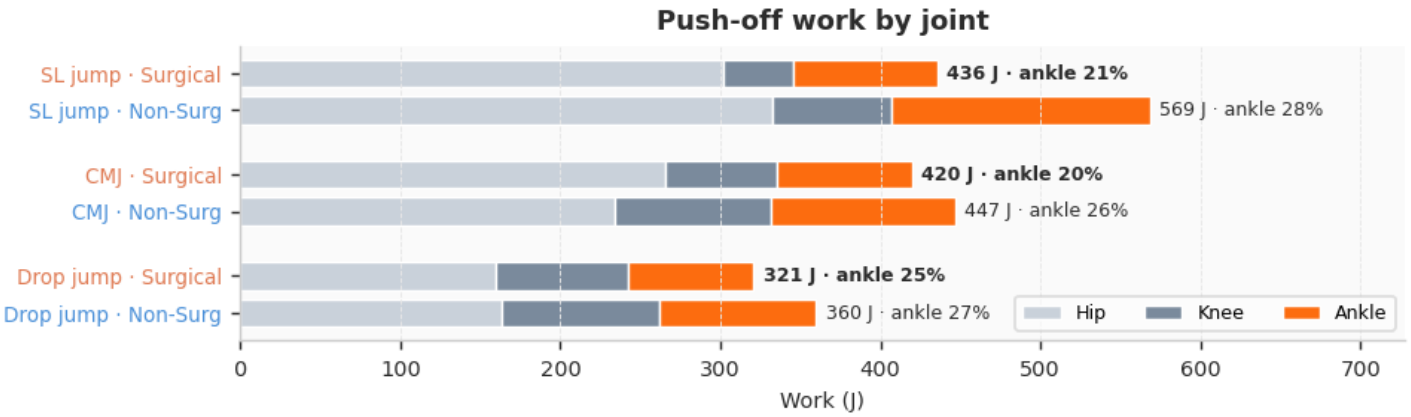
Fast loading is avoided. Drop jump RFD 67% and loading rate 69% of the non-surgical side, against peak force 79%. The surgical leg accepts load slowly — the Achilles' job is the opposite.

Not a reactive jump. Drop jump contact is 0.48 s (target < 0.25 s); RSI 1.26 m/s comes from height, not from a fast stretch-shortening rebound.

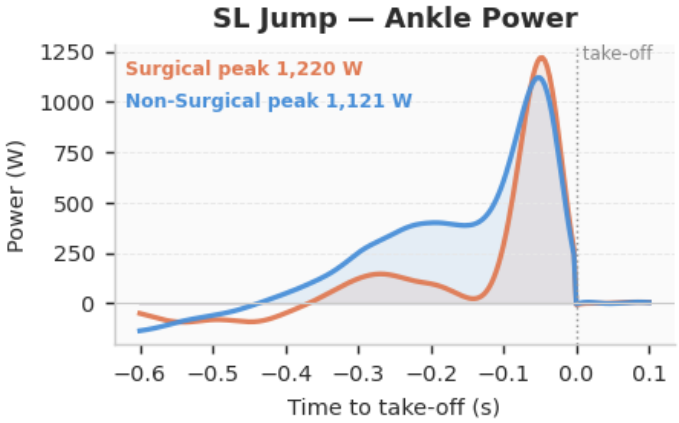
Landing bypasses the ankle. The surgical ankle absorbs 53% of the landing energy the other side does, while the surgical hip absorbs 129%.

Balance failed. The surgical limb could not hold eyes-closed single-leg stance in 2 of 2 conditions — and the non-surgical limb failed too, so this is a bilateral deficit, not only an asymmetry.

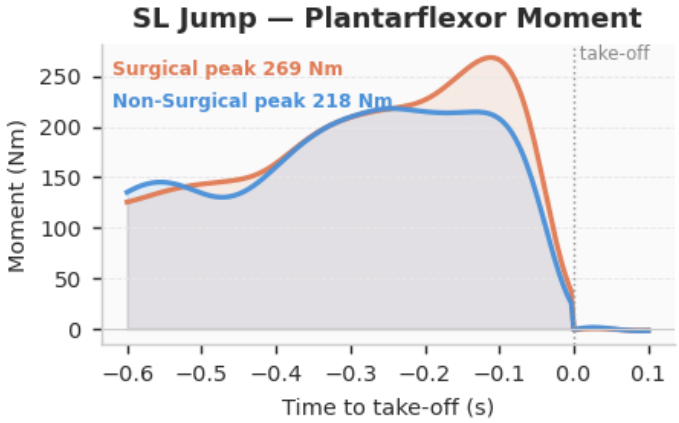
Hop distance hides it. Triple hop is 97% — horizontal hopping can be driven by the hip, so it doesn't clear the calf.



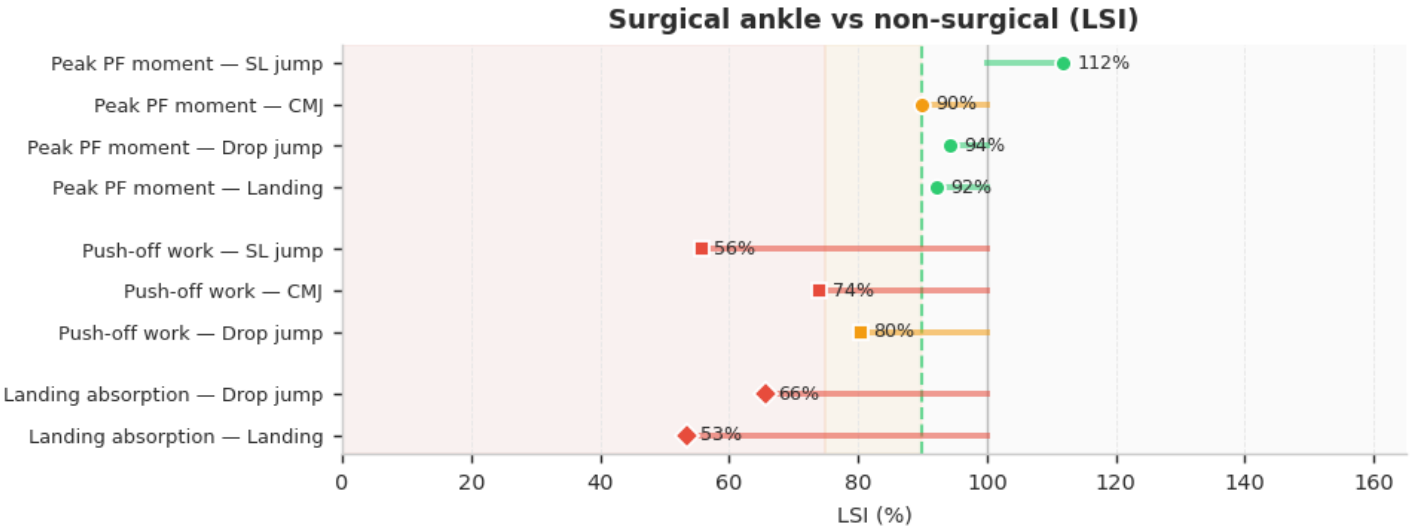
Push-off (positive) joint work per leg. Ankle share in orange — a smaller ankle share on the surgical side with a bigger hip share = the hip is doing the calf's job.



Ankle power through push-off. Area above zero = push-off work.



Plantarflexor moment. Similar height but less power = slow calf.



Each line runs from 100% (symmetric) to the surgical ankle's value. Peak moment near 100% with push-off work far below it = the calf has strength but not speed.

PRIORITIES FOR THE PT

1. **Add a single-leg heel raise to failure** at the next visit (reps and heel height each leg). It is the standard Achilles return criterion and isn't in this score yet.
2. **Calf speed, not just strength.** Peak plantarflexor moment is already symmetric, so the gap is how fast the calf can produce it: fast concentric single-leg heel raises, loaded calf jumps and pogo-style work, keeping heavy calf loading in to hold capacity.
3. **Reactive progression.** Bilateral pogos → single-leg pogos → short-contact drop jumps, cueing a stiff ankle and contact under 0.25 s. Re-test RSI and contact time.
4. **Landing through the forefoot.** Low-box forefoot landings and decelerations so the surgical calf takes its share of the landing instead of the hip.
5. **Single-leg balance.** Eyes-closed single-leg stance on firm ground, then foam, both limbs — the non-surgical side needs it too if it also failed.

DROP JUMP

ABOUT THIS TEST

The Drop Jump is a reactive bilateral task: the athlete drops from a box and immediately jumps as high and as quickly as possible. It loads the calf–Achilles unit through a fast stretch-shortening cycle — the same demand as sprinting and cutting. After Achilles repair the common compensations are a longer ground contact, a slower and lower initial force rise on the surgical limb (reduced RFD / loading rate), a heel-biased landing, and shifting work to the knee and hip. The Reactive Strength Index (jump height ÷ contact time) summarises how well the athlete uses elastic energy; contact time itself shows whether the jump was truly reactive (< 0.25 s) or a slow, guarded rebound.

Metric	Surgical	Non-Surg	LSI / Δ
Peak GRF (N)	1311.8	1659.9	79.0%
Impulse (N·s)	464.1	552.4	84.0%
RFD 100ms (N/s)	19666.6	29483.9	66.7%
Loading Rate 50ms (N/s)	17307.7	25071.5	69.0%
Landing Sym 200ms (%) *(bilateral)*	—	—	73.1%
Impact Transient (N)	815.7	900.2	90.6%
Time-to-Peak (ms)	167.4	70.7	—
Jump Height (in)	23.9	—	—
Contact Time (s)	0.482	—	—
RSI (m/s)	1.26	—	—
Peak Knee Flexion (°)	85.9	91.4	93.9%
Peak Knee Valgus (°)	8.2	3.0	Δ +5.2°
Peak Tibial IR (°)	-6.4	-0.4	Δ -6.0°
Tibial Rot Excursion (°)	6.4	2.7	Δ +3.7°
Peak Hip Flexion (°)	79.6	82.9	Δ -3.2°
Peak Hip Adduction (°)	-2.8	-0.4	Δ -2.4°
Peak Hip IR (°)	11.7	4.4	Δ +7.3°
Peak Ankle Dorsiflexion (°)	26.4	29.5	Δ -3.1°
Peak Plantarflexor Moment (Nm)	174.8	185.2	94.4%
Peak Knee Ext Moment (Nm)	143.6	197.8	72.6%
AP COP Excursion (mm)	116.9	134.0	—
Trials analysed	3	—	—
Peak Power (W/kg)	71.8	—	—
Limb Peak Power (W)	3333.0	3576.4	93.2%

Limb Push-off Work (J)	536.2	580.2	92.4%
Implied Drop Height (cm)	29.0	—	—
Ankle Absorption (J)	50.1	76.4	65.6%
Knee Absorption (J)	62.3	119.9	51.9%
Hip Absorption (J)	98.4	118.6	83.0%
Ankle share of absorption (%)	23.2	24.1	—
Ankle +Work (J)	78.8	98.0	80.4%
Knee +Work (J)	82.5	98.3	84.0%
Hip +Work (J)	159.5	163.6	97.5%
Ankle share of +work (%)	24.4	27.1	—

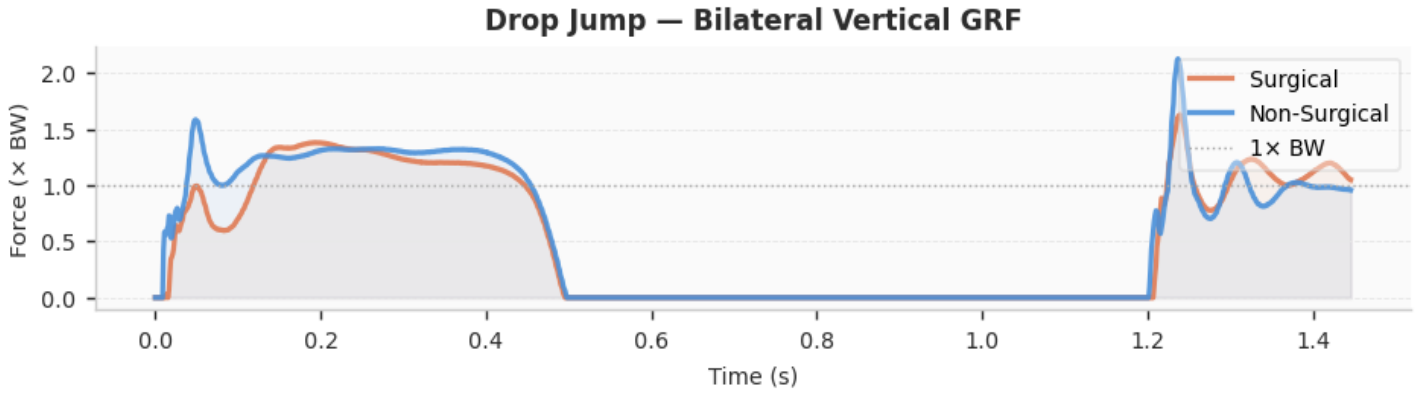
CLINICAL INTERPRETATION

RSI 1.26 m/s with 0.48 s contact (⚠ below RTS target ≥ 1.30). Long contact / low reactive strength — the surgical calf–Achilles unit is not yet tolerating fast stretch-shortening loads. Progress pogo → short-contact drop jumps.

Contact time 0.48 s — not a reactive (fast stretch-shortening) contact; target < 0.25 s.

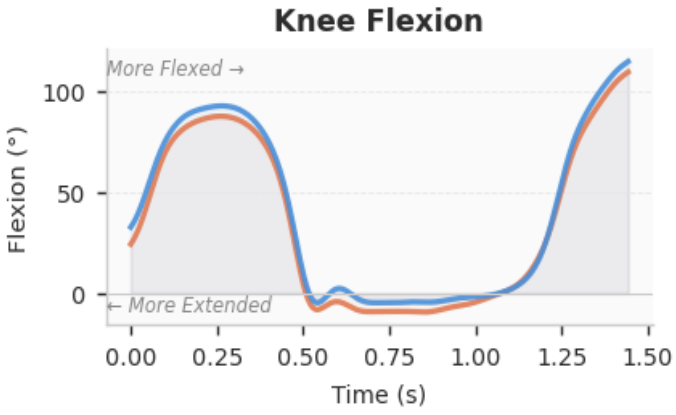
Landing symmetry 73% LSI: Surgical limb loading 27% less than non-surgical — surgical guarding detected.

Targets: RSI ≥ 1.30 m/s · Contact < 0.25 s · RFD & load rate LSI ≥ 85–90% · Landing LSI 90–110%.

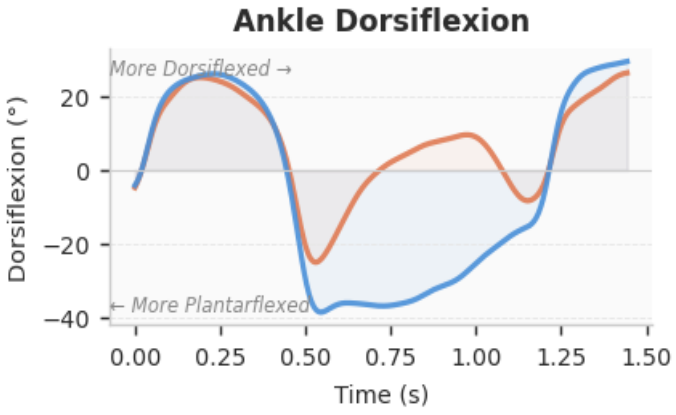


Bilateral GRF (normalised to BW). A slower, lower initial rise on the surgical side = reduced fast loading through the calf–Achilles unit.

Landing symmetry 73% at 200ms — surgical limb offloading detected.

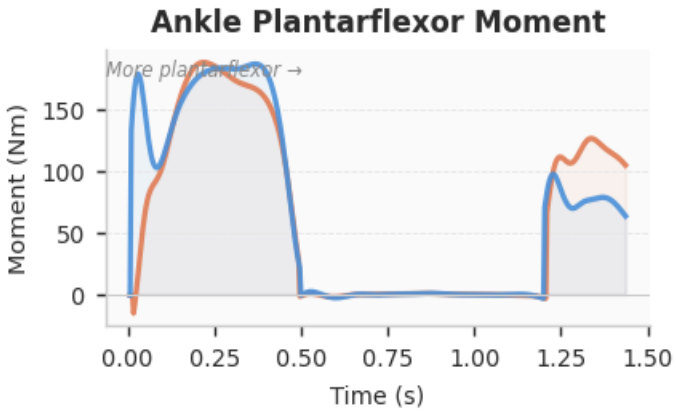


Knee flexion. Deeper knee bend on landing can be used to offload the ankle.

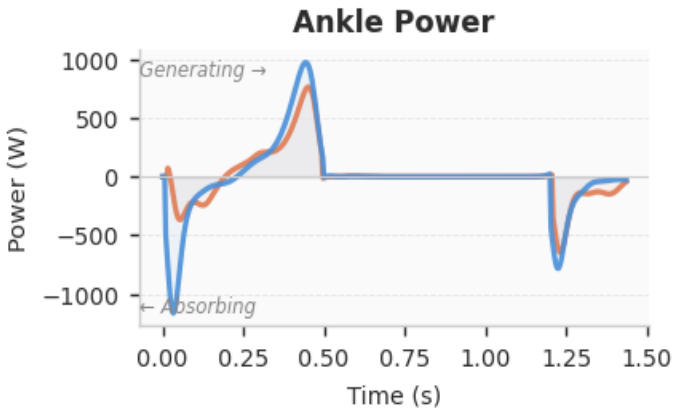


Ankle dorsiflexion. Reduced surgical dorsiflexion under load = calf guarding; excess = tendon elongation / reduced stiffness.

***Left panel:** Surgical limb knee flexion depth near-symmetric (LSI 94%) — approaching return-to-sport target; minor deficit remains.*



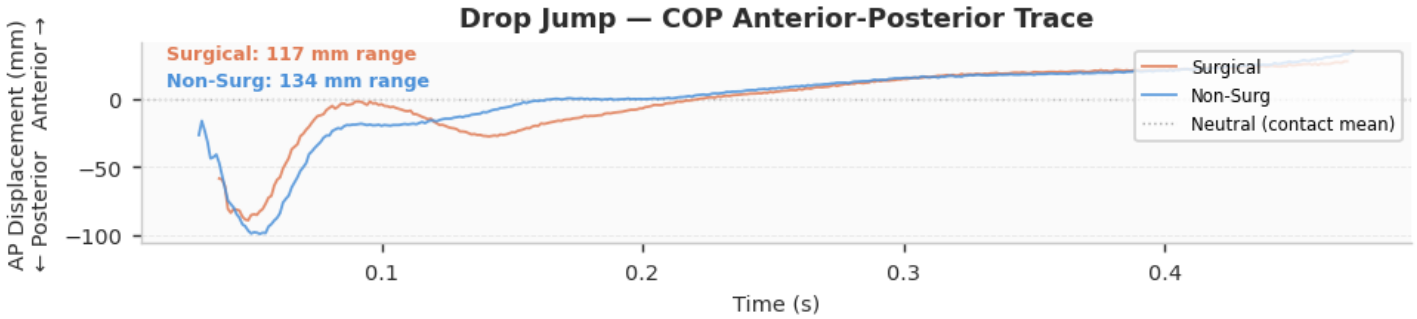
Plantarflexor moment through landing and rebound — how hard the calf–Achilles unit is loaded.



Ankle power: below zero = calf absorbing the landing, above zero = calf driving the rebound.

Left panel: *Surgical limb peak plantarflexor moment near-symmetric (LSI 94%) — approaching return-to-sport target; minor deficit remains.*

Right panel: *Surgical limb ankle push-off work shows a moderate asymmetry (LSI 80%) — continued loading progression recommended before clearance.*



AP COP during the first contact (zero-centred). Posterior / heel bias on the surgical side = reduced forefoot loading through the Achilles.

AP COP excursion: surgical 117 mm vs non-surgical 134 mm — less fore-aft travel on the surgical side — guarded or restricted loading.

Surgical limb bilateral landing symmetry demonstrates a significant deficit (LSI 73%) — targeted strengthening and re-testing indicated.

SINGLE-LEG VERTICAL JUMP

ABOUT THIS TEST

The Single-Leg Vertical Jump isolates each limb so the uninvolved side cannot compensate. Final push-off depends on the plantarflexors, so after Achilles repair single-leg height is one of the more sensitive power measures. Compare jump height with net impulse: if impulse is near-symmetric but height is not, the athlete is producing force slowly over a longer contact rather than quickly through the calf.

Metric	Surgical	Non-Surg	LSI / Δ
Jump Height (in)	11.9	13.9	85.7%
Peak GRF (N)	2026.3	1869.0	108.4%
Net Impulse (N·s)	233.6	252.3	92.6%
Peak RFD (N/s)	9492.0	10472.1	—
Contact Time (s)	0.96	0.89	—
Peak Knee Flexion (°)	54.0	69.1	—
Peak Knee Valgus (°)	2.5	7.3	Δ -4.7°
Peak Tibial IR (°)	-2.3	3.1	Δ -5.4°
Peak Ankle Dorsiflexion (°)	22.0	30.9	Δ -8.8°
Peak Plantarflexor Moment (Nm)	264.5	236.3	111.9%
Peak Hip Adduction (°)	8.2	11.1	Δ -3.0°
Peak Knee Ext Moment (Nm)	161.5	231.0	69.9%
AP COP Excursion (mm)	94.9	154.6	—
Trials analysed	3	3	—
Peak Power (W/kg)	45.2	47.7	94.8%
Push-off Work (J)	769.5	978.4	78.6%
Peak RPD (W/s)	28221.2	23610.2	119.5%
Ankle +Work (J)	90.0	161.7	55.7%
Knee +Work (J)	44.1	74.6	59.1%
Hip +Work (J)	301.5	332.2	90.8%
Ankle share of +work (%)	20.7	28.5	—

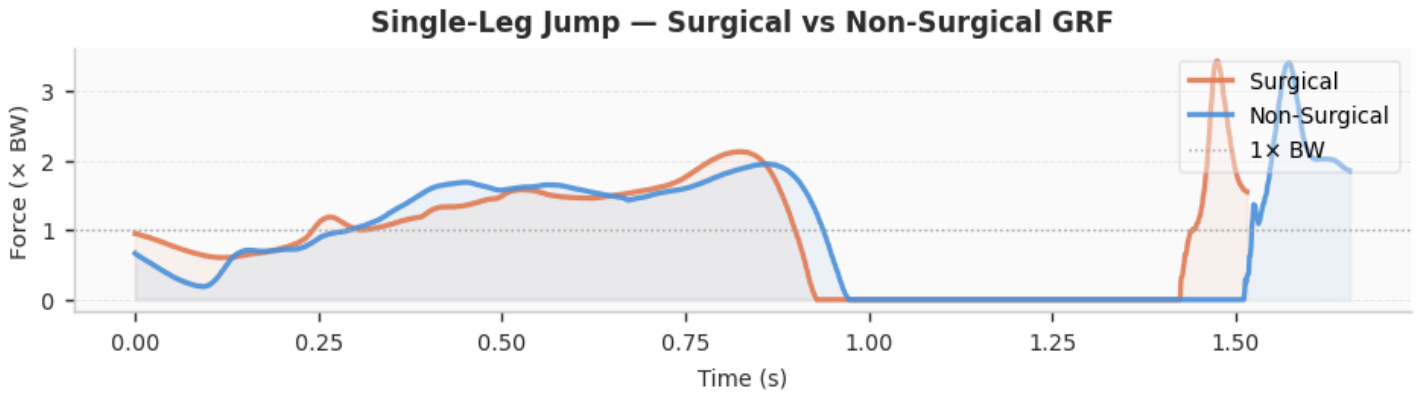
CLINICAL INTERPRETATION

Isolates each limb — the surgical calf must produce the final push-off without help.

Jump height LSI 86%: ⚠ 14% below non-surgical. Continued plyometric loading required.

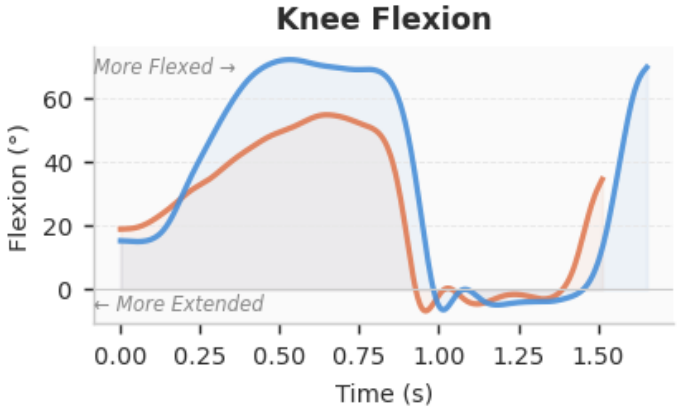
Ankle +work LSI 56% vs hip 91%: The surgical ankle does far less push-off work than the other side while the hip makes up the difference — a hip-dominant strategy that protects the calf.

Targets: Jump height LSI ≥ 90% · Net impulse LSI ≥ 90% · Heel-raise LSI ≥ 90%.

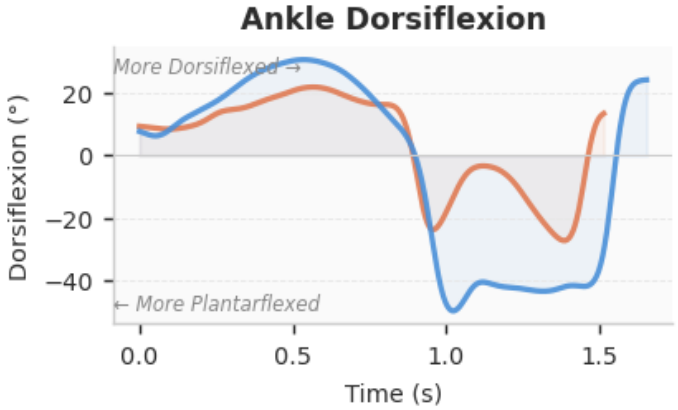


Per-limb GRF (separate trials). A later, drawn-out surgical peak = slow push-off.

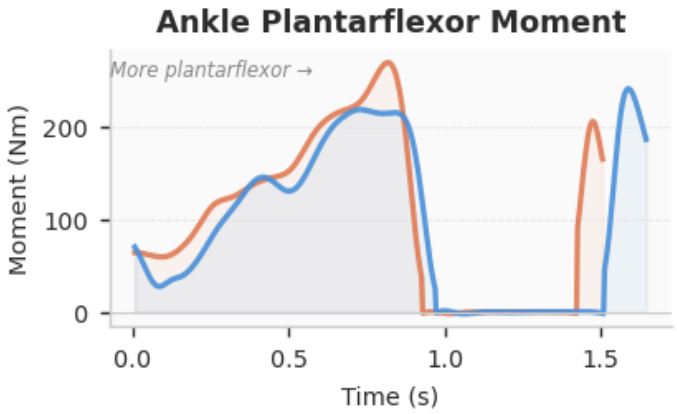
Surgical limb single-leg jump height shows a moderate asymmetry (LSI 86%) — continued loading progression recommended before clearance.



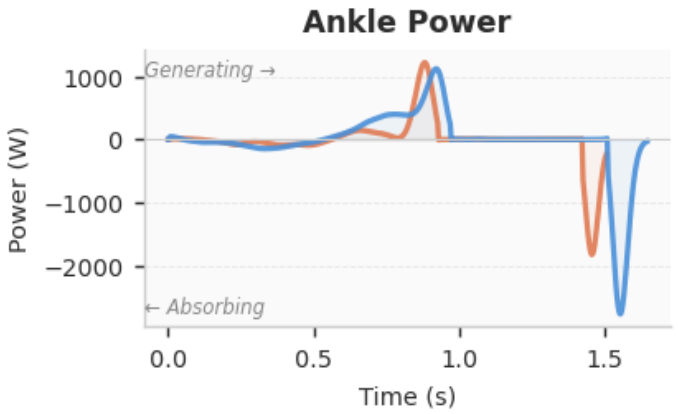
Knee flexion per limb. Reduced surgical depth = shortened, guarded countermovement.



Ankle dorsiflexion per limb — plantarflexion range at push-off drives single-leg height.



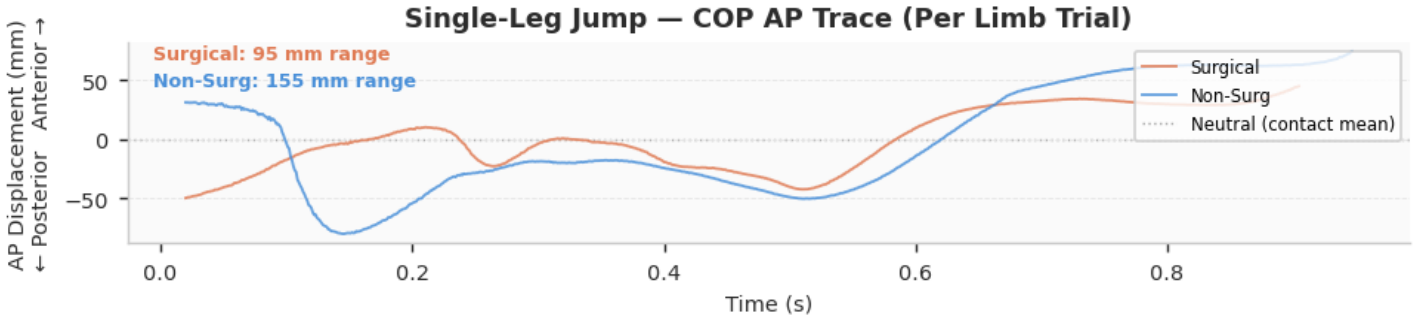
Plantarflexor moment per limb. A similar peak with less power = slow calf.



Ankle power per limb — the positive burst before take-off is calf push-off work.

Left panel: *Surgical limb peak plantarflexor moment is symmetric with the non-surgical side (LSI 112%) — meets return-to-sport threshold.*

Right panel: *Surgical limb ankle push-off work demonstrates a significant deficit (LSI 56%) — targeted strengthening and re-testing indicated.*



AP COP during the take-off contact (zero-centred). Less anterior travel = limited push-off through the forefoot.

AP COP excursion: surgical 95 mm vs non-surgical 155 mm — less fore-aft travel on the surgical side — guarded or restricted loading.

Surgical limb single-leg jump height shows a moderate asymmetry (LSI 86%) — continued loading progression recommended before clearance.

MAX VERTICAL JUMP (CMJ)

ABOUT THIS TEST

The Countermovement Jump is a bilateral power test with a relatively long, slow stretch-shortening cycle. Because the hip and knee can contribute most of the work, CMJ symmetry often returns before the calf has recovered — a symmetric CMJ does not clear an Achilles. Propulsive impulse and peak force LSI show whether the surgical limb is contributing to push-off.

Metric	Surgical	Non-Surg	LSI / Δ
Peak Propulsive Force (N)	1216.9	1296.9	93.8%
Net Propulsive Impulse (N·s)	147.9	189.1	78.2%
Peak Braking Force (N)	1217.9	1276.0	95.4%
Jump Height (in)	25.0	—	—
Flight Time (s)	0.719	—	—
Unweighting Impulse Surg/NS	82.7	79.2	—
Peak Knee Flexion (°)	84.1	88.0	95.6%
Peak Knee Valgus (°)	10.1	2.3	Δ +7.8°
Peak Tibial IR (°)	-4.8	2.7	Δ -7.5°
Tibial Rot Excursion (°)	7.5	6.1	Δ +1.4°
Peak Hip Flexion (°)	96.7	98.0	Δ -1.4°
Peak Hip Adduction (°)	-3.1	2.2	Δ -5.3°
Peak Hip IR (°)	11.4	1.8	Δ +9.6°
Peak Ankle Dorsiflexion (°)	28.0	29.6	Δ -1.6°
Peak Plantarflexor Moment (Nm)	163.2	181.4	90.0%
Peak Knee Ext Moment (Nm)	122.7	157.0	78.1%
AP COP Excursion (mm)	92.8	13.0	—
Trials analysed	2	—	—
Braking Impulse (N·s)	49.5	74.3	66.6%
Peak Power (W/kg)	78.7	—	—
Limb Peak Power (W)	3659.7	3915.2	93.5%
Limb Push-off Work (J)	559.9	636.5	88.0%

Ankle +Work (J)	85.0	114.8	74.1%
Knee +Work (J)	69.2	98.4	70.3%
Hip +Work (J)	265.7	233.6	113.7%
Ankle share of +work (%)	20.3	25.7	—

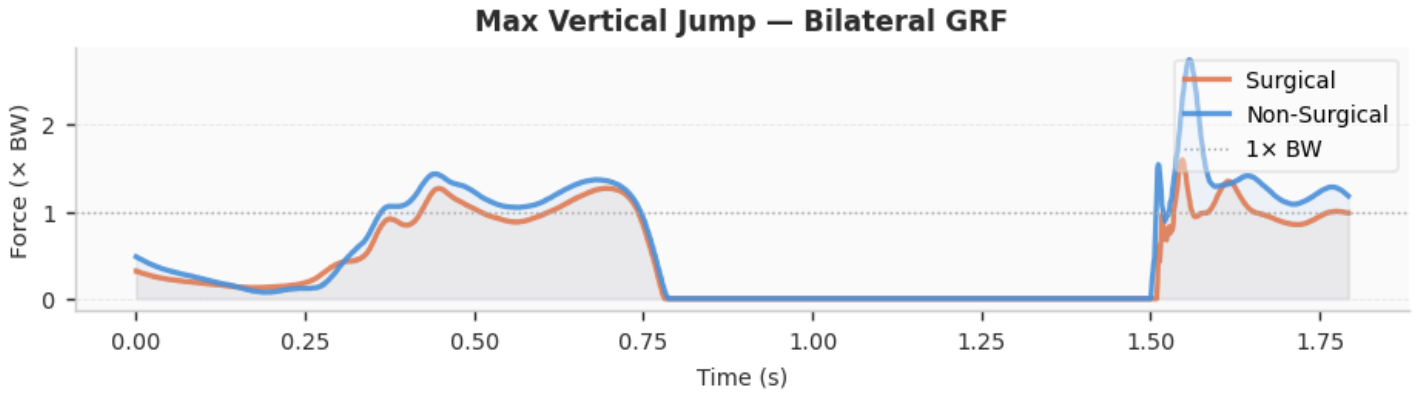
CLINICAL INTERPRETATION

Bilateral CMJ — slow stretch-shortening; usually recovers before fast-SSC tasks after Achilles repair.

Braking force LSI 95%: Symmetric bilateral output.

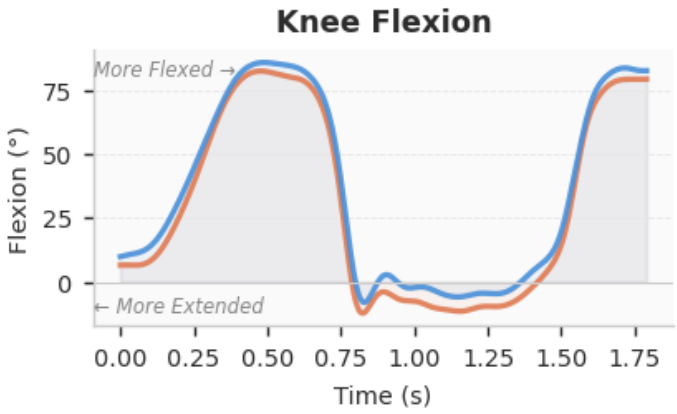
Propulsion LSI 78%. Targets: ≥ 90% propulsion, ≥ 90% peak force.

Ankle +work LSI 74% vs hip 114%: The surgical ankle contributes less push-off work while the hip makes up the difference — the calf is being bypassed in the propulsive phase.

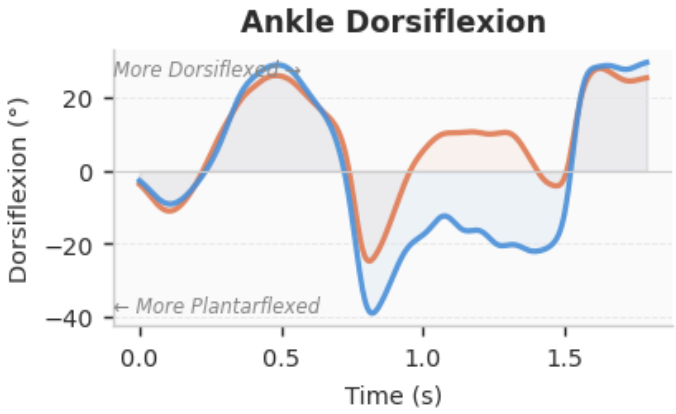


Full CMJ: unweighting → eccentric → propulsion → flight. Propulsive peaks should be symmetric.

Surgical limb propulsive impulse shows a moderate asymmetry (LSI 78%) — continued loading progression recommended before clearance.

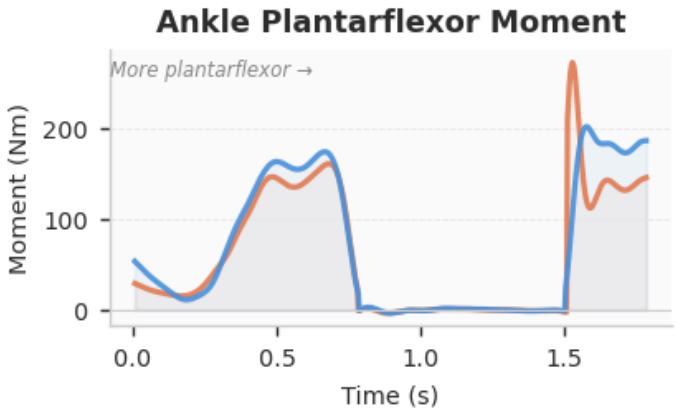


Knee flexion depth. Asymmetric countermovement depth = compensatory loading strategy.

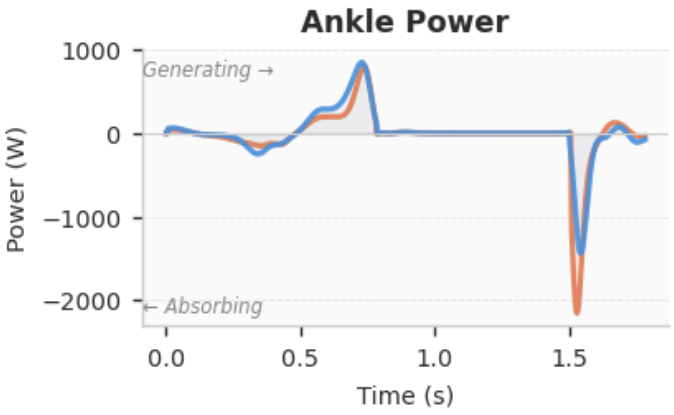


Ankle dorsiflexion — the final push-off through plantarflexion drives jump height.

***Left panel:** Surgical limb countermovement knee flexion depth is symmetric with the non-surgical side (LSI 96%) — meets return-to-sport threshold.*



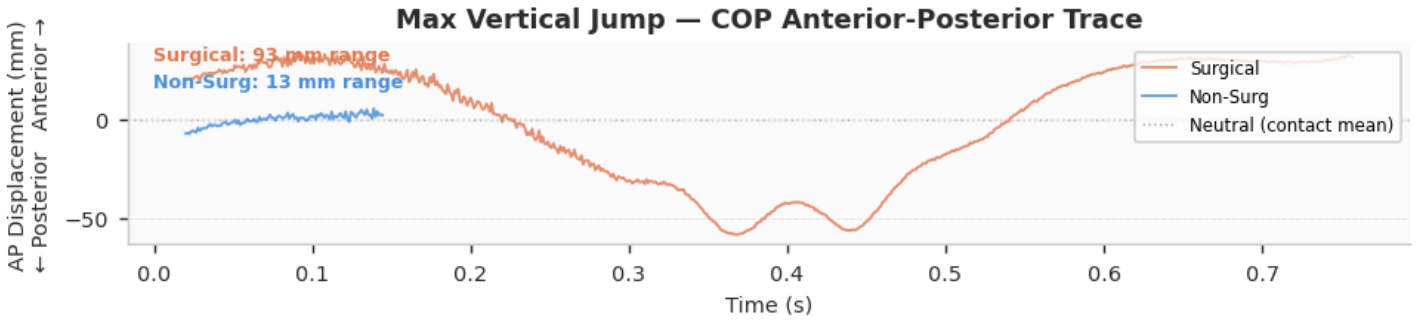
Plantarflexor moment through the countermovement and push-off.



Ankle power: the late positive burst before take-off is the calf's push-off.

Left panel: *Surgical limb peak plantarflexor moment shows a moderate asymmetry (LSI 90%) — continued loading progression recommended before clearance.*

Right panel: *Surgical limb ankle push-off work demonstrates a significant deficit (LSI 74%) — targeted strengthening and re-testing indicated.*



AP COP during the propulsive contact (zero-centred). Less anterior travel on the surgical side = reduced push-off through the forefoot.

AP COP excursion: surgical 93 mm vs non-surgical 13 mm — more fore-aft travel on the surgical side — less controlled foot loading.

Surgical limb peak propulsive force is symmetric with the non-surgical side (LSI 95%) — meets return-to-sport threshold.

DROP LANDING

ABOUT THIS TEST

The Drop Landing isolates load acceptance: the athlete drops from a box and sticks the landing without a re-jump. After Achilles repair the surgical limb often lands heel-first and accepts load more slowly, which shows up as a lower loading rate and impact transient rather than a lower peak force. Symmetric loading rate means the calf is eccentrically controlling the forefoot-to-heel transition on both sides.

Metric	Surgical	Non-Surg	LSI / Δ
Peak GRF (N)	1339.1	1456.6	91.9%
Impulse (N·s)	621.7	640.8	97.0%
Loading Rate 50ms (N/s)	23242.3	22571.8	103.0%
Impact Transient (N)	708.4	1043.4	67.9%
Peak Knee Flexion (°)	111.1	117.3	94.8%
Peak Knee Valgus (°)	10.8	4.2	Δ +6.6°
Peak Tibial IR (°)	-4.6	1.0	Δ -5.5°
Tibial Rot Excursion (°)	5.4	4.3	Δ +1.1°
Peak Hip Flexion (°)	106.1	105.6	Δ +0.5°
Peak Hip Adduction (°)	-5.8	-7.8	Δ +2.0°
Peak Hip IR (°)	16.7	8.9	Δ +7.8°
Peak Ankle Dorsiflexion (°)	21.2	27.8	Δ -6.6°
Peak Plantarflexor Moment (Nm)	135.9	147.3	92.3%
Peak Knee Ext Moment (Nm)	102.0	137.2	74.3%
AP COP Excursion (mm)	73.3	106.4	—
Trials analysed	3	—	—
Ankle Absorption (J)	49.6	92.8	53.5%
Knee Absorption (J)	81.1	145.4	55.8%
Hip Absorption (J)	209.3	162.1	129.1%
Ankle share of absorption (%)	14.7	23.3	—

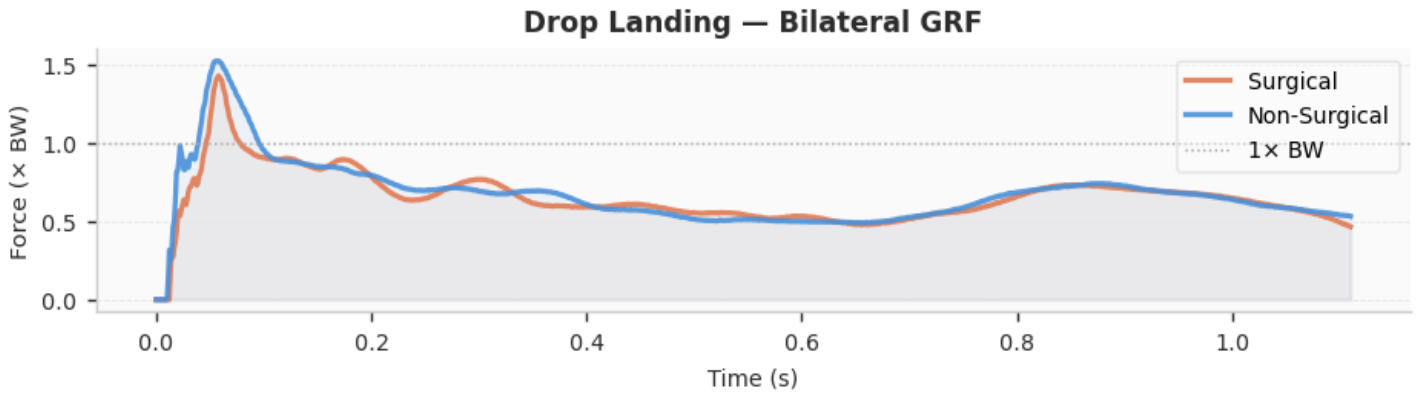
CLINICAL INTERPRETATION

Absorptive landing — assesses how quickly and symmetrically the surgical limb accepts load.

Load rate LSI 103%: Symmetric loading strategy.

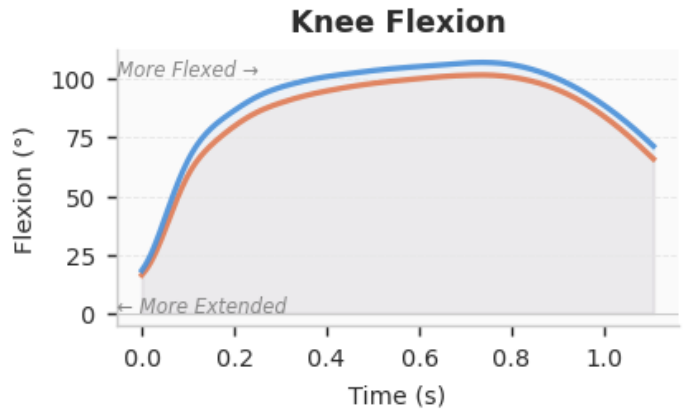
Impact transient LSI 68% — surgical limb initial impact 32% lower.

Targets: Load rate LSI ≥ 85–90% · Impact transient LSI 90–110% · Peak GRF LSI ≥ 90%.

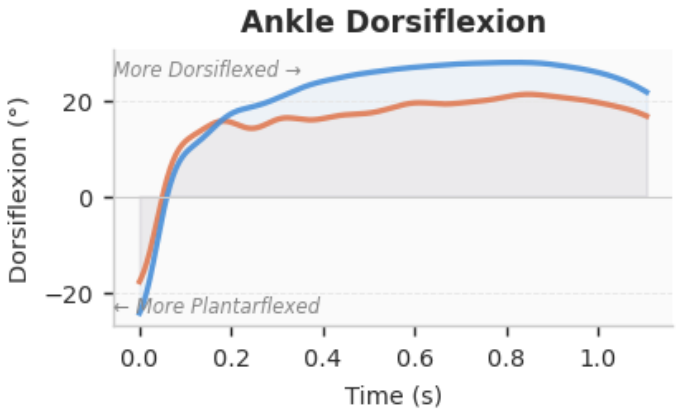


Bilateral GRF. Lower, later initial peak on the surgical side = slower load acceptance.

Peak GRF symmetry 92% — approaching return-to-sport target; minor deficit remains.

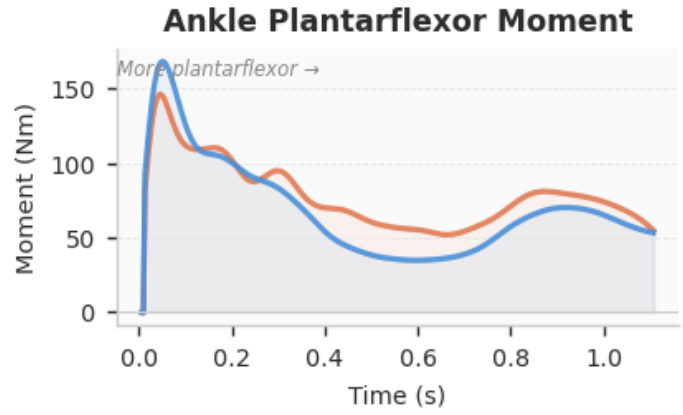


Knee flexion at landing. Extra knee bend can substitute for ankle absorption.

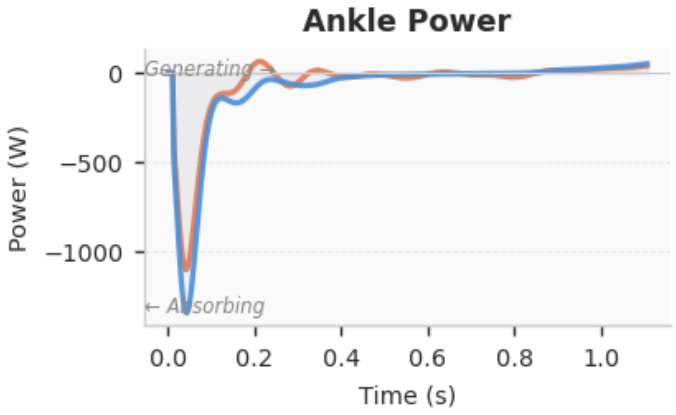


Ankle dorsiflexion at landing — eccentric calf control.

***Left panel:** Surgical limb landing knee flexion depth near-symmetric (LSI 95%) — approaching return-to-sport target; minor deficit remains.*



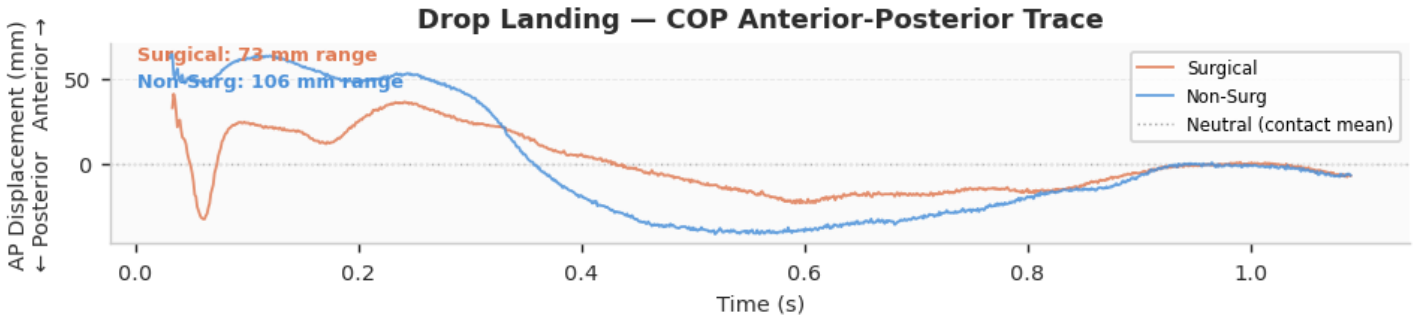
Plantarflexor moment during landing — eccentric calf loading.



Ankle power during landing: the area below zero is the energy the calf absorbs.

***Left panel:** Surgical limb peak plantarflexor moment near-symmetric (LSI 92%) — approaching return-to-sport target; minor deficit remains.*

***Right panel:** Surgical limb ankle landing absorption demonstrates a significant deficit (LSI 53%) — targeted strengthening and re-testing indicated.*



AP COP during landing contact (zero-centred). Posterior bias on surgical side = heel-dominant / reduced forefoot loading.

AP COP excursion: surgical 73 mm vs non-surgical 106 mm — less fore-aft travel on the surgical side — guarded or restricted loading.

Surgical limb peak landing force absorption near-symmetric (LSI 92%) — approaching return-to-sport target; minor deficit remains.

BALANCE — STANDARD (EYES CLOSED, FIRM SURFACE)

ABOUT THIS TEST

Eyes-closed single-leg stance removes visual feedback, so balance depends on ankle and foot proprioception and on the calf's ability to make fast small corrections. Higher COP velocity means more corrective activity. After Achilles repair, reduced calf strength and altered tendon stiffness commonly increase sway on the surgical limb. Limb symmetry (LSI ≥ 85–90%) is the primary criterion.

Metric	Surgical	Non-Surg	LSI / Δ
Stance hold	FAIL	FAIL	—
COP Mean Velocity (mm/s)	380.0	140.5	37.0%
COP Ellipse Area (mm²)	40703.1	3873.8	9.5%
Total Excursion (mm)	2712.6	614.8	—
AP Range (mm)	168.6	106.0	—
ML Range (mm)	366.4	45.0	—
RMS Displacement (mm)	74.7	22.0	—

CLINICAL INTERPRETATION

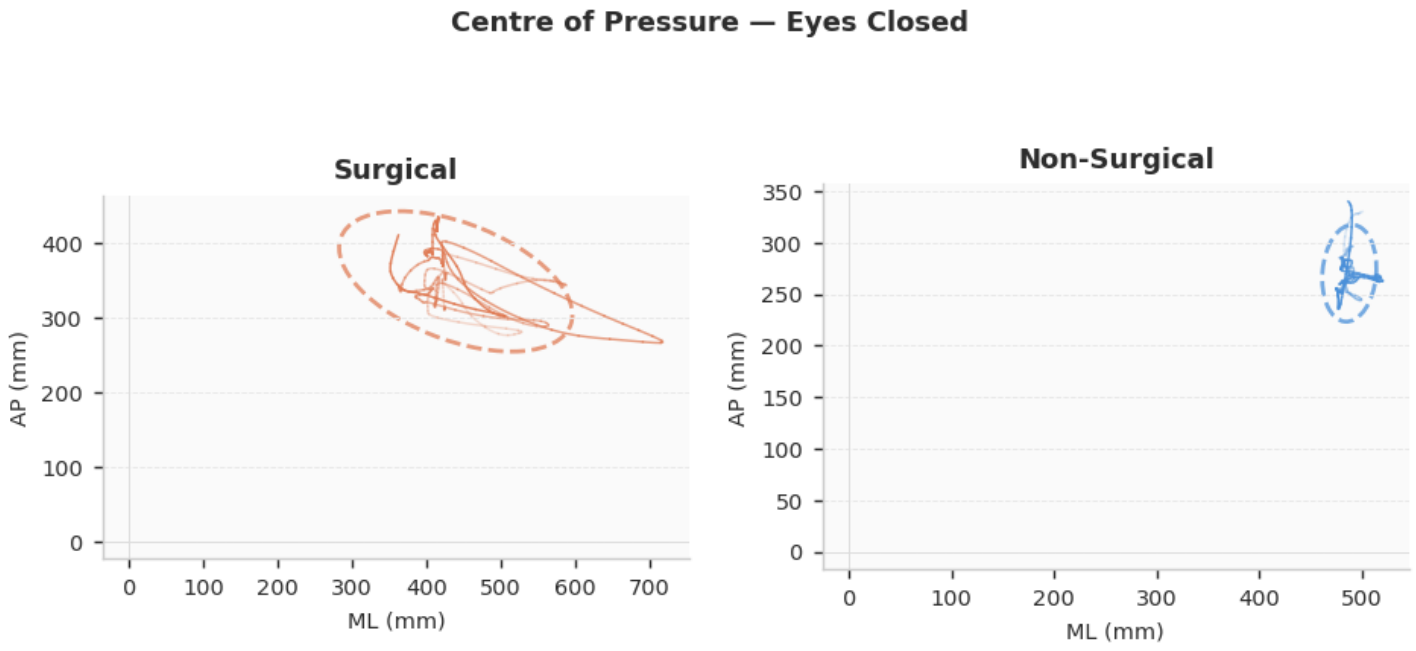
Eyes-closed single-leg stance on firm surface. COP velocity is the most sensitive metric — higher speed = more corrective activity = reduced proprioception.

Surgical 380 mm/s vs Non-Surgical 140 mm/s — Surgical limb less stable. Proprioceptive retraining indicated.

COP Velocity LSI 37% (NS÷Surg×100; below 100% = surgical side sways more). LSI target ≥ 90%. COP low-pass filtered at 10 Hz.

✗ **Failed stance hold:** surgical limb — foot moved + touchdown; non-surgical limb — touchdown. The athlete could not hold single-leg eyes-closed stance without moving the foot or touching down; this counts as a fail (sway values below include the loss of balance).

Surgical limb failed: velocity symmetry is capped at 50% for scoring — the non-surgical limb also failed, so this is a bilateral balance deficit, not just an asymmetry.



COP trajectory (10 Hz low-pass) — eyes closed. Tighter ellipse = steadier stance. 95% confidence ellipse shown.

BALANCE — AIREX PAD (EYES CLOSED, UNSTABLE SURFACE)

ABOUT THIS TEST

Eyes-closed single-leg stance removes visual feedback, so balance depends on ankle and foot proprioception and on the calf's ability to make fast small corrections. Higher COP velocity means more corrective activity. After Achilles repair, reduced calf strength and altered tendon stiffness commonly increase sway on the surgical limb. Limb symmetry (LSI ≥ 85–90%) is the primary criterion.

Metric	Surgical	Non-Surg	LSI / Δ
Stance hold	FAIL	FAIL	—
COP Mean Velocity (mm/s)	315.0	225.1	**50.0%**
COP Ellipse Area (mm²)	33657.9	17728.4	**52.7%**
Total Excursion (mm)	3634.9	2893.3	—
AP Range (mm)	195.1	191.3	—
ML Range (mm)	221.2	137.2	—
RMS Displacement (mm)	62.8	44.4	—

CLINICAL INTERPRETATION

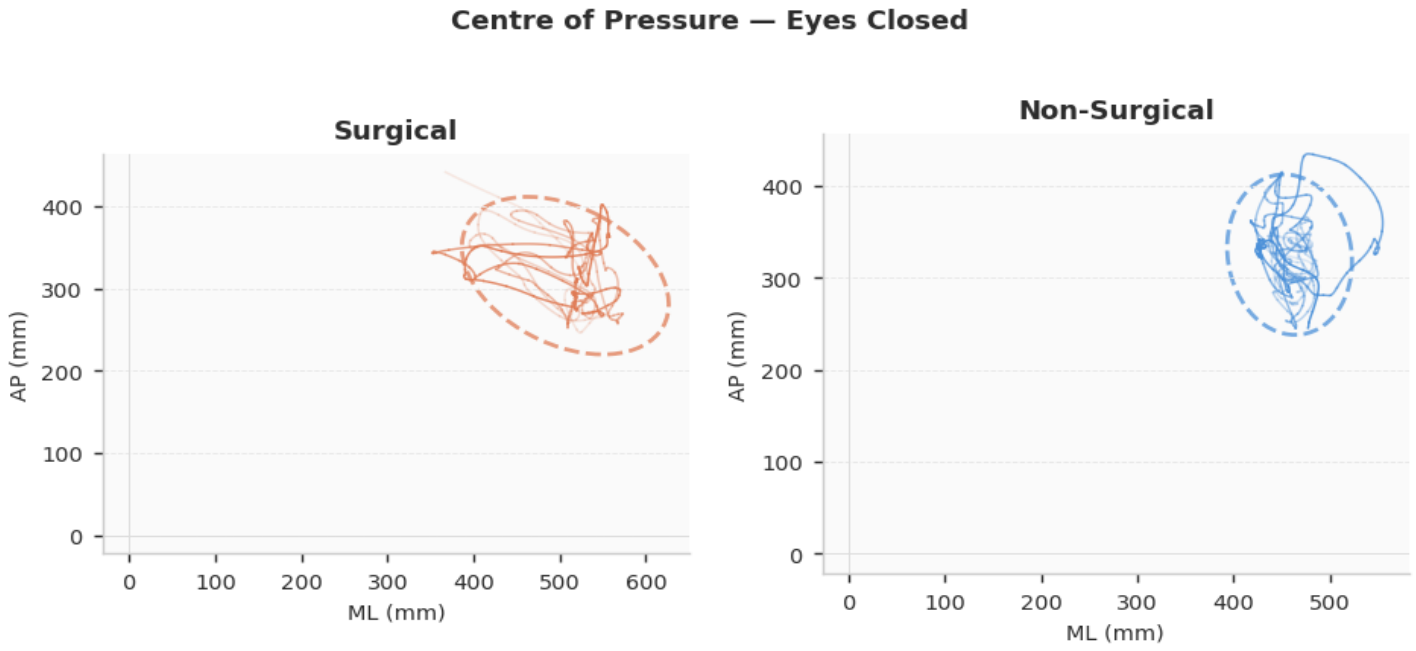
Eyes-closed single-leg stance on Airex foam pad. Higher COP values are expected — the unstable surface challenges proprioceptive and neuromuscular systems. Symmetry between limbs (LSI ≥ 85%) is the key target.

Surgical 315 mm/s vs Non-Surgical 225 mm/s — Surgical limb less stable. Proprioceptive retraining indicated.

COP Velocity LSI 50% (NS÷Surg×100; below 100% = surgical side sways more). LSI target ≥ 90%. COP low-pass filtered at 10 Hz.

✗ **Failed stance hold:** surgical limb — foot moved; non-surgical limb — foot moved. The athlete could not hold single-leg eyes-closed stance without moving the foot or touching down; this counts as a fail (sway values below include the loss of balance).

Surgical limb failed: velocity symmetry is capped at 50% for scoring — the non-surgical limb also failed, so this is a bilateral balance deficit, not just an asymmetry.



COP trajectory (10 Hz low-pass) — eyes closed. Tighter ellipse = steadier stance. 95% confidence ellipse shown.

ENDURANCE SQUAT (30s)

ABOUT THIS TEST

The 30-Second Endurance Squat assesses limb symmetry under sustained bilateral loading. It is knee- and hip-dominant, so for an Achilles repair it mainly checks for whole-limb offloading under fatigue; calf endurance is assessed with the single-leg heel raise to failure.

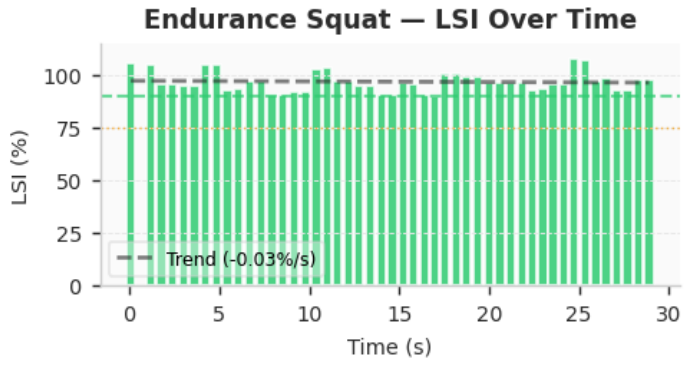
Metric	Surgical	Non-Surg	LSI / Δ
Mean Peak Force LSI (%) *(bilateral)*	—	—	97.1%
LSI First 10s (%) *(bilateral)*	—	—	97.0%
LSI Last 10s (%) *(bilateral)*	—	—	97.4%
Fatigue Drift (Δ%)	0.4	—	—
Total Reps Detected	48.0	—	—
Mean Peak Force Surg (N)	903.3	—	—
Mean Peak Force NS (N)	931.9	—	—
AP COP Excursion — Surg / NS (mm)	218.2	225.8	—
Peak Knee Ext Moment (Nm)	123.6	143.1	86.4%

CLINICAL INTERPRETATION

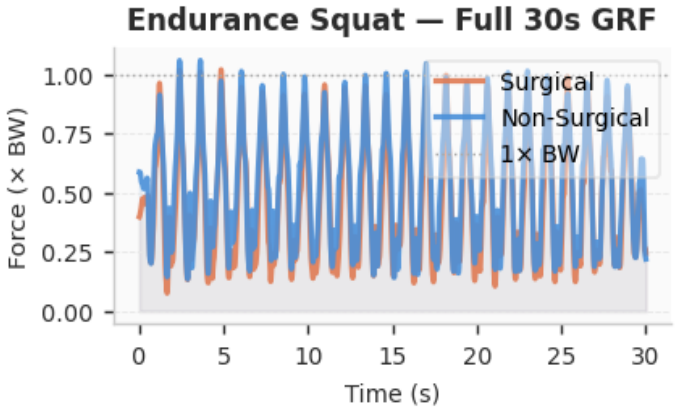
30s bilateral squat — knee-dominant endurance; does not directly load the calf.

Fatigue drift 0.4%: ✓ Symmetry maintained under fatigue.

Targets: Mean LSI ≥ 90% · Fatigue drift < 5%. Pair with the heel-raise test for calf endurance.

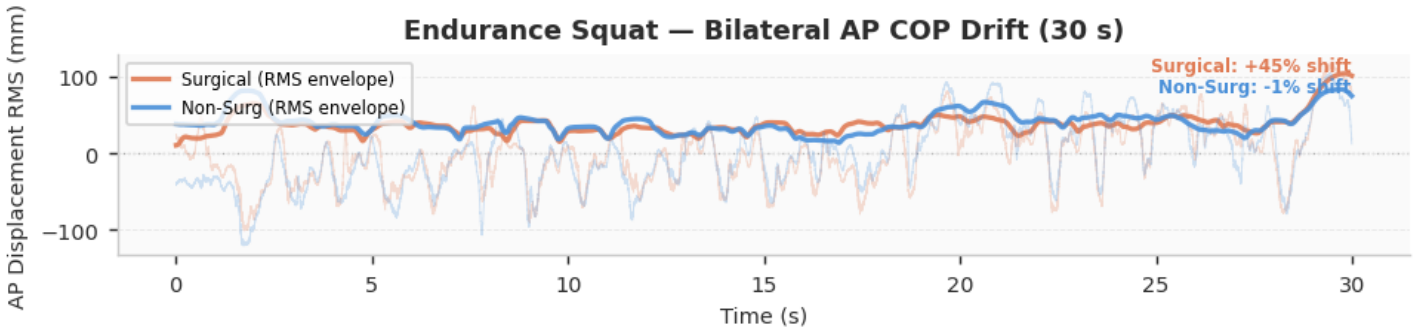


Per-rep LSI across 30s. Declining trend = neuromuscular fatigue breakdown.



Full 30s GRF trace. Force magnitude and rhythm across the set.

***Left panel:** Surgical limb mean 30s force symmetry is symmetric with the non-surgical side (LSI 97%) — meets return-to-sport threshold.*



Bilateral AP COP drift across 30s (zero-centred, 1s RMS envelope). Surgical-side amplitude shrinking or trending posterior = offloading under fatigue. Diverging traces = progressive asymmetry with sustained effort.

Fatigue drift: 0.4% LSI change across 30s — symmetry maintained under sustained effort.

Surgical limb mean force symmetry over 30 s is symmetric with the non-surgical side (LSI 97%) — meets return-to-sport threshold.